

1. Solid
2. Chemical change $\because$ propane reacts with oxygen to produce carbon dioxide and water.
3. 879000000 W
4. Ni
5. H_2CrO_4
6. Compound $\because$ it contains sulfur and oxygen atoms bonded in a fixed ratio.
7. $10 \frac{\text{mol}}{\text{L}}$
8. 0.0043 m
9. No new substance is formed $\because$ Water evaporates but remains water in the vapor phase.
10. $7.7 \times 10^{-1} \text{ m}$
11. SiF_4
12. $7.08 \times 10^1 \text{ Gm}$
13. $3.089 \times 10^3 \frac{\text{kg}}{\text{m}^3}$
14. Heterogeneous mixture $\because$ it is a combination of two or more distinct substances that are not uniformly distributed.
15. Br
16. $5.79 \times 10^{-1} \text{ m}$
17. $3.284 \times 10^0 \text{ kg}$
18. 3.4% $\because$ % Error = $\left| \frac{E-A}{A} \right| \times 100\%$
19. 18.0g
20. 1.6 mm
21. 470.5 g
22. 0.000000000000064 s
23. 0.00000000000028 g
24. $5.172 \frac{\text{m}}{\text{s}}$
25. 9.383 g
26. 5°C
27. Solid
28. $8.7 \times 10^0 \text{ mm}$
29. Intensive property. Color is an intensive property $\because$ the color of a substance does not depend on the size of the sample.
30. Pure substances: A, B $\because$ the mass ratio of E_1 to E_2 in A and B is approximately an integer multiple of the ratio in pure substance X. When two elements form different pure compounds, the masses of one element that combine with a fixed mass of the other are related by integer ratios (Law of Multiple Proportions).
31. $0.47 \frac{\text{g}}{\text{mL}}$
32. $8.54 \times 10^1 \text{ fm}$
33. 0.000000000000024 s
34. 718.28 nm
35. Arsenic acid
36. $2.6970 \times 10^2 \text{ nm}$
37. 0.000066 m
38. Barium most commonly forms the cation Ba^{2+} .
39. F_2O is a molecular compound.
40. 138.08 cm
41. Chemical property. The ability to rust of iron is a chemical property; it describes iron's tendency to react with oxygen and water

to form iron oxide, a new substance.

42. 8.549×10^0 km
43. Sample B $\therefore \rho_A = 0.90 \frac{\text{g}}{\text{mL}}$ and $\rho_B = 2.7 \frac{\text{g}}{\text{mL}}$
44. Cl_4Si is a molecular compound.
45. 2.80×10^1 cm
46. X is a pure substance; Mass of E_1 : 6.2 g. X is a pure substance $\therefore$ the ratio of the masses of E_1 to E_2 is the same for all four samples. A pure substance always has a constant composition. If $m_{E1} + m_{E2} = 10.0$ g, and you have calculated the mass fraction $m_{E1}/(m_{E1} + m_{E2})$ from the samples, you would expect m_{E1} to be 6.2 g.
47. Compound
48. Mixture. It is a (heterogeneous) mixture; it contains water, sugars, acids, pulp, and other compounds in varying proportions.
49. 38.5 g
50. Sulfuric acid
51. It is a homogeneous mixture. $\therefore$ it is a combination of two or more substances with a uniform composition and appearance.
52. Silver chloride: AgCl
53. Mixture. It is a (likely heterogeneous) mixture; it contains minerals, organic matter, air, and water in varying proportions.
54. No, chlorine and iodine cannot form an ionic compound.
55. 23 protons and 28 neutrons.
56. 268 K
57. Yes, chromium and iodine form an ionic compound with the empirical formula CrI_3 .
58. Homogeneous mixture $\therefore$ it is a combination of two or more substances with a uniform composition and appearance.
59. 364.55 K
60. 2.61 mL
61. 4.7×10^{-8} m
62. 47.16 mL
63. Heterogeneous mixture $\therefore$ it is a combination of two or more distinct substances that are not uniformly distributed.
64. 8.362 km
65. 3.0 mL
66. 9.046×10^1 g
67. chlorine monofluoride
68. 0.653 m
69. Mercury
70. Barium sulfide: BaS
71. Physical change $\therefore$ water changes phase from liquid to solid but remains H_2O .
72. 50°C
73. Chloric acid
74. 3.975×10^1 MPa
75. I is a neutral atom. It has 53 protons and 53 electrons.
76. 2.25×10^{11} Hz
77. 9.02×10^8 W
78. 3.0% $\therefore \% \text{ Error} = \left| \frac{E-A}{A} \right| \times 100\%$
79. P_4O_{10}
80. 51.2 μg
81. Rubidium hypobromite: RbBrO
82. $5. \times 10^3$ mol
83. Uranium

84. $0.30 \frac{\text{g}}{\text{mL}}$
85. Strontium phosphate decahydrate: $\text{Sr}_3(\text{PO}_4)_2 \cdot 10\text{H}_2\text{O}$
86. $3.6 \times 10^{-5} \text{ m}$
87. 0.64 g
88. Physical property. The melting point of ice is a physical property; it can be observed without changing the chemical identity of water.
89. $0.58 \frac{\text{g}}{\text{mL}}$
90. $1.00 \times 10^1 \text{ fm}$
91. Ba^{2+} is a cation. It has 56 protons and 54 electrons.
92. 53 cm
93. Yes, a new substance is formed $\therefore$ Carbon dioxide gas is produced by the chemical reaction.
94. Random error $\therefore$ the measurements fluctuate unpredictably in both directions.
95. $595.3 \frac{\text{kg}}{\text{m}^3}$
96. Barium oxide dihydrate: $\text{BaO} \cdot 2\text{H}_2\text{O}$
97. The SI base unit for time is the second (symbol: s).
98. $9.53 \times 10^{11} \text{ Hz}$
99. Iron (iii) oxide heptahydrate: $\text{Fe}_2\text{O}_3 \cdot 7\text{H}_2\text{O}$
100. $5.9923 \times 10^3 \text{ g}$
101. Ammonium oxide: $(\text{NH}_4)_2\text{O}$
102. Compound $\therefore$ it contains hydrogen, nitrogen, and oxygen atoms bonded in a fixed ratio.
103. $72.2 \mu\text{g}$
104. $7.549 \times 10^1 \text{ g}$
105. $8.26 \times 10^1 \text{ Gm}$
106. 0.0026 m
107. $23 \text{ }^\circ\text{F}$
108. X is a mixture $\therefore$ the ratio of the masses of E_1 to E_2 is not the same in all four samples. Mixtures do not have a fixed composition, so different samples can contain different proportions of the two elements.
109. Physical change $\therefore$ the shape changes temporarily but no new substance forms.
110. 3.927 g
111. 0.000000020 m
112. $6.0 \times 10^{-1} \text{ m}$
113. $4.4482 \times 10^3 \text{ g}$
114. Ammonium carbonate: $(\text{NH}_4)_2\text{CO}_3$
115. $20 \frac{\text{mol}}{\text{L}}$
116. None of the substances are pure substances. $\therefore$ none of the mass ratios of E_1 to E_2 are integer multiples of the E_1 to E_2 ratio in pure substance X. When two elements form different pure compounds, the masses of one element that combine with a fixed mass of the other are related by integer ratios (Law of Multiple Proportions).
117. Systematic error $\therefore$ every measurement is consistently too high by the same amount due to a calibration flaw.
118. $59 \text{ }^\circ\text{F}$
119. diiodine pentoxide
120. 273.35 K
121. $0.91 \frac{\text{g}}{\text{mL}}$
122. 27 protons and 32 neutrons.
123. $1 \mu\text{g} = 10^{-6} \text{ g}$
124. $10 \text{ }^\circ\text{C}$

125. 46 mL
126. 8.03 g
127. The SI base unit for amount of substance is the mole (symbol: mol).
128. No new substance is formed $\therefore$ The wire changes shape but remains copper.
129. 44.9 fm
130. 0.59% $\therefore$ % Error = $\left| \frac{E-A}{A} \right| \times 100\%$
131. Pure substance. It is a pure substance (element); it contains only mercury atoms and has a fixed, uniform composition.
132. 1.5×10^0 mm
133. Ag_2O is an ionic compound.
134. HNO_3
135. H_3AsO_4
136. 5.4×10^0 mg
137. $0.50 \frac{\text{g}}{\text{mL}}$
138. CO_2
139. 6.498 km
140. 357.35 K
141. Element $\therefore$ it contains only phosphorus atoms.
142. Chlorine most commonly forms the anion Cl^- .
143. $3.00 \times 10^1 \mu\text{g}$
144. $30 \frac{\text{mol}}{\text{L}}$
145. 2.429×10^0 g
146. Systematic error $\therefore$ the incorrect technique consistently shifts every measurement in the same direction.
147. 2.4×10^3 mol
148. Caesium phosphate: Cs_3PO_4
149. Ni
150. 29.3 cm
151. 796.20 nm
152. 0.000035 m
153. 6765.9 m
154. 240 mol
155. 94 protons and 145 neutrons.
156. 8.5×10^{-3} m
157. oxygen difluoride
158. Sample A $\therefore \rho_A = 0.997 \frac{\text{g}}{\text{mL}}$ and $\rho_B = 0.565 \frac{\text{g}}{\text{mL}}$
159. $6.9 \frac{\text{m}}{\text{s}}$
160. Extensive property. Particle count is an extensive property $\therefore$ the number of particles depends on the size of the sample.
161. 253 K
162. 6.3×10^0 mg
163. The SI base unit for mass is the kilogram (symbol: kg).
164. Element
165. Pure substances: B, C $\therefore$ the mass ratio of E_1 to E_2 in B and C is approximately an integer multiple of the ratio in pure substance X. When two elements form different pure compounds, the masses of one element that combine with a fixed mass of the other are related by integer ratios (Law of Multiple Proportions).
166. Phosphorus most commonly forms the anion P^{3-} .
167. 0.692 m

168. 5.966 kg
169. 7.529×10^1 MPa
170. 1335.9 m
171. 3.1×10^0 mg
172. It is an element. $\therefore$ it contains only one element and is pure in its composition.
173. 95 °F
174. 0.75 m
175. 82.0 Gm
176. $9.456 \times 10^3 \frac{\text{kg}}{\text{m}^3}$
177. 3.649×10^1 g
178. Chemical property. The ability to rust of iron is a chemical property; it describes iron's tendency to react with oxygen and water to form iron oxide, a new substance.
179. 39.1 cm
180. 527000000000 Hz
181. Yes, potassium and iodine form an ionic compound with the empirical formula KI.
182. 5504.8 m
183. 1.0×10^{-11} g
184. 10^{-6}
185. Sample B $\therefore \rho_A = 0.99 \frac{\text{g}}{\text{mL}}$ and $\rho_B = 1.85 \frac{\text{g}}{\text{mL}}$
186. K^+ is a cation. It has 19 protons and 18 electrons.
187. 4.4 mL
188. 0.000000000000097 s
189. Gallium
190. 21.28 MPa
191. Intensive property. Temperature is an intensive property $\therefore$ the temperature of a sample does not depend on its quantity.
192. Gas
193. $13.8 \frac{\text{m}}{\text{s}}$
194. 0.000000000078 g
195. X is a mixture $\therefore$ the ratio of the masses of E_1 to E_2 is not the same in all four samples. Mixtures do not have a fixed composition, so different samples can contain different proportions of the two elements.
196. It is an element. $\therefore$ it contains only one element and is pure in its composition.
197. 278 K
198. 0.000000078 m
199. 4.6066×10^3 g
200. Element